EXAMPLE 2.6

The resistance of a wire is 20 Ω. What will be new resistance, if it is stretched uniformly 8 times its original length?

Solution

$R_1 = 20\ \Omega$, $R_2 = ?$

Let the original length of the wire (l_1) be l.

New length, $l_2 = 8l_1$ *(i.,e)* $l_2 = 8l$

Original resistance, $R_1 = \rho\dfrac{l_1}{A_1}$

New resistance $R_2 = \rho\dfrac{l_2}{A_2} = \dfrac{\rho(8l)}{A_2}$

Though the wire is stretched, its volume remains unchanged.

Initial volume = Final volume

$A_1 l_1 = A_2 l_2, \qquad A_1 l = A_2(8l)$

$$\frac{A_1}{A_2} = \frac{8l}{l} = 8$$

By dividing equation for R_2 by equation for R_1, we get

$$\frac{R_2}{R_1} = \frac{\rho(8l)}{A_2} \times \frac{A_1}{\rho l}$$

$$\frac{R_2}{R_1} = \frac{A_1}{A_2} \times 8$$

Substituting the value of $\dfrac{A_1}{A_2}$, we get

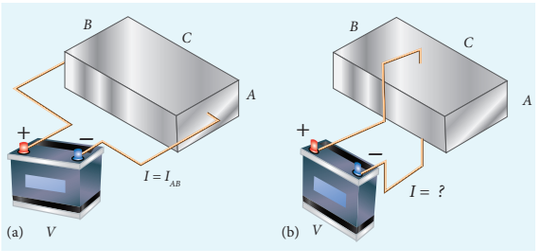
$$\frac{R_2}{R_1} = 8 \times 8 = 64$$

$$R_2 = 64 \times 20 = 1280\ \Omega$$

Hence, stretching the length of the wire has increased its resistance.

EXAMPLE 2.7

Consider a rectangular block of metal of height A, width B and length C as shown in the figure.



If a potential difference of V is applied between the two faces A and B of the block (figure (a)), the current I_{AB} is observed. Find the current that flows if the same potential difference V is applied between the two faces B and C of the block (figure (b)). Give your answers in terms of I_{AB}.

Solution

In the first case, the resistance of the block

$$R_{AB} = \rho\frac{\text{length}}{\text{Area}} = \rho\frac{\text{C}}{\text{AB}}$$

The current $I_{AB} = \dfrac{V}{R_{AB}} = \dfrac{V}{\rho}\cdot\dfrac{AB}{C}$ (1)

In the second case, the resistance of the block $R_{BC} = \rho\dfrac{A}{\text{BC}}$

The current $I_{BC} = \dfrac{V}{R_{BC}} = \dfrac{V}{\rho}\cdot\dfrac{BC}{A}$ (2)

To express I_{BC} interms of I_{AB}, we multiply and divide equation (2) by AC, we get

$$I_{BC} = \frac{V}{\rho}\cdot\frac{BC}{A}\frac{AC}{AC} = \left(\frac{V}{\rho}\cdot\frac{AB}{C}\right)\cdot\frac{C^2}{A^2} = \frac{C^2}{A^2}\cdot I_{AB}$$

Since $C > A$, the current $I_{BC} > I_{AB}$



The human body contains a large amount of water which has low resistance of around 200 Ω and the dry skin has high resistance of around 500 k Ω. But when the skin is wet, the resistance is reduced to around 1000 Ω. This is the reason why repairing the electrical connection with the wet skin is always dangerous.

12th Physics_Vol-1_Unit 2_English Medium.indd 89 12-12-2022 17:22:12